Addendum (Cybersecurity)

to

Installation Manual for VRS 002 G2 and VRS 003

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1 Scope and purpose

This document guides the application of the IEC 61162-460 ind IACS UR E27 guidelines to the VRS 002 G3 and VRS 003 (in the following, both abbreviated as VRS) for it to be installed in a 460-Network as a certified 460-Gateway.

It provides all the relevant information system integrators need to successfully apply an E27 and 460-compliant configuration and the content needed for defining the network infrastructure and topology.

1.1 References

[1] MAN13534 Manual for Installation manual for Vessel Remote Server, VRS 002 G2 and VRS 003

[2] Maritime navigation and radiocommunication equipment and systems – Digital interfaces – Part 460: Multiple talkers and multiple listeners – Ethernet interconnection – Safety and security (IEC 61162-460:2024)

[3] IACS UR E27 - Cyber resilience of on-board systems and equipment.

[4] Maritime navigation and radiocommunication equipment and systems – Digital interfaces – Part 450: Multiple talkers and multiple listeners – Ethernet connection (IEC61162-450:2024)

1.2 Terms and Abbreviations

API	Application Programming Interface
Controlled network	A controlled network is any network designed to operate such that it does not pose any security risks to any of the connected network nodes, as defined in [2].
DHCP	Dynamic Host Configuration Protocol
DVA	Digital Video Analogue Streaming
HMI	Human-Machine Interface
IACS	International Association of Classification Societies
JSON	JavaScript Object Notation
RDI	Remote Data Interface
SFI	System Function ID
System Integrator	Person or organization responsible for the functionality of the integrated 460-network
TPM	Trusted Platform Module
Uncontrolled network	Any network (including the internet) that is not IEC 61162-450, IEC 61162-460 compliant, or a controlled network.
UR	Unified Requirements
VRS	Vessel Remote Server
VDR	Voyage Data Recorder
WAN	Wide Area Network

2 System Overview

The VRS is certified as a 460-Gateway against IEC 61162-460:2024 ed. 3 ([2]), as well as IACS UR E27 [3]

Prior to installing the VRS, a network topology drawing should be consulted to define which interfaces are to be connected to controlled and uncontrolled networks since the VRS acts as the gateway between controlled and uncontrolled networks. This input is needed when setting up the VRS as described in 3.2.

In a controlled network, access is restricted, and network traffic is monitored according to the specifications of all nodes, so no undisclosed data is present.
For example, a VDR should be connected to a network port that is a part of the controlled network on the VRS. The ethernet connection to the VDR must not traverse switches,ers, or other equipment that could be considered uncontrolled, e.g., network equipment that is also connected to other general-purpose networks on the ship.

Some of the available services provided by the VRS are unsuitable for use in uncontrolled networks. The security notifications in the web configuration interface ensures that all services not allowed exposure to an uncontrolled network are turned off or only enabled on controlled networks.

2.1 No direct connections

The VRS does not implement the full feature set for a 460-Gateway as described in [2].
Direct Connections are connections between controlled (e.g., a 460-Node) and uncontrolled networks directly through the VRS.

All network traffic traversing the VRS, is initiated and terminated by the VRS itself. Direct Connections are not possible, as such, the management of Direct Connections is not implemented or available in the VRS.

2.2 Network monitoring function (IEC 61162-460 only)

The VRS does not perform the functionality of a Network Monitor as described in [2]. It does, however, fulfill the requirements of a client and interfaces to a Network Monitoring Function already available in the network.

The VRS reports to the Network Monitor with required network-related information via remote Syslog.
Note that the VRS only reports on the outbound traffic flows, as seen from the perspective of the VRS.

An example of the output from the VRS when monitoring is enabled and a report is sent to the network monitoring function can be seen in

Appendix A.
The complete list of content as shown, is transmitted every time a change is done to the network on the VRS (not more often than once per 1 minute) or every 30 minutes. The full content must be filtered for potential changes by the monitoring function.

This is strictly only mandatory for a VRS deployed as a 460-Gateway, in other cases the functionality can be disabled and the VRS will still be compliant to IACS UR E27.

2.3 System management function (IEC 61162-460 only)

The VRS does not perform the functionality of the System Management function as described in [2]. It does, however, fulfill the requirements of a client to the System Management Function and provides appropriate APIs to the System Management Function.

2.3.1 Snapshots

A snapshot is a backup of the currently running configuration of a particular VRS, it is specific to the VRS and cannot be downloaded but is only stored on the VRS itself.

Snapshots can be managed with a custom API endpoint on the VRS. A snapshot is created either manually or automatically on every boot. The System Management Function can utilize this API to reinstate the VRS to previous settings.

Create snapshot:

```
curl -d "" -ksu <username>:<password> -X POST 'https://<vrs-ip-address>/vrs/api/snapshots'
```

Verify the new snapshot is created:

```
curl -ksu <username>:<password> -X GET 'https://<vrs-ip-address>/vrs/api/snapshots’
```

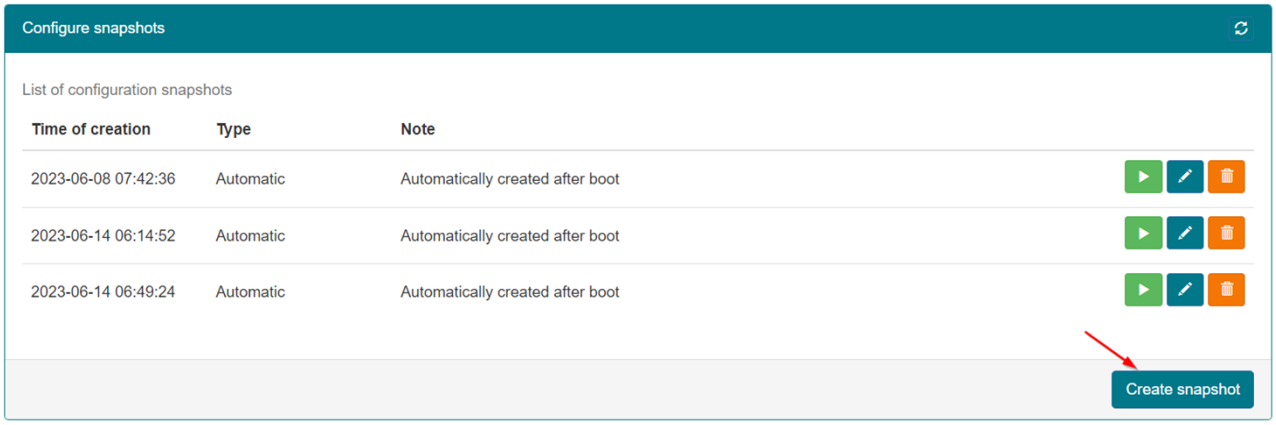
A list of information regarding the snapshots is returned as a JSON object and contains the following:

create_time	Time of creation
flag	Reserved for future use
id	Unique identifier (used when restoring a snapshot)
note	Custom note related to the snapshot
type	Either 1 (Manual) or 2 (Automatic)

Reinstate a previous snapshot:

```
curl -d "" --insecure --location --request POST 'https://<vrs-ip-
address>/vrs/api/snapshots/restore/<snapshot-id>?auth-user=<username>&auth-
pwd=<password>&rescue=false'
```

Snapshots can also be created and restored with the configuration interface by going to ***System -> Snapshots***.


Configure snapshots
List of configuration snapshots
Time of creation
Type
Note
2023-06-08 07:42:36
Automatic
Automatically created after boot
2023-06-14 06:14:52
Automatic
Automatically created after boot
2023-06-14 06:49:24
Automatic
Automatically created after boot
Create snapshot

2.3.2 Configuration backup/restore

In addition to Snapshots stored locally on the VRS, it is also possible to download a maintenance backup to apply it later. These backups are not specific to the VRS but can also be applied on a replacement VRS.

Download backup:

```
curl -sku <username>:<password> 'https://<vrs-ip-address>/vrs/download/backup' -output <file-name>.tar.gz
```

Upload backup:

```
curl -k -X POST -F 'file=@<file-name>.tar.gz' -u <username>:<password> 'https://<vrs-ip-address>/vrs/api/restore
```

2.4 Redundancy

The VRS is not considered a critical component and should not be placed on any critical path.

The redundancy requirements defined in [2] are not applicable to the VRS.

3 Installation

3.1 Feature summary

The VRS serves as an aggregation point for agnostic data collection on board the vessel. Data is received, processed and repackaged by the VRS and subsequently sent towards endpoints locally on the ship or to the Danelec cloud on shore.

The VRS is not intended to be used by the crew onboard the vessel, but it is important to note that a Cloud-enabled VRS will always maintain a secure connection to the Cloud for remote management and monitoring.

A VRS in both Gateway and Cloud-enabled mode is placed on the boundary between the trusted and untrusted networks, if networks connected to the VRS are to be isolated, the VRS must be manually turned off, see 4.4.

3.2 Mark network interfaces as controlled

A crucial step in properly setting up the VRS is configuring which interfaces are controlled.

Interfaces 2, 3, and 4 are marked as controlled in the default configuration.
Due to security concerns, some services should not be enabled towards uncontrolled networks (see 3.3.2).
The VRS firmware monitors the enabled services, and provides a security notice if an offending configuration change is made.

For example, a VDR or RDI connection should come from a controlled network where no insecure network nodes are present and data is properly segregated.

Interface 1 (usually used to configure the VRS) is uncontrolled since any machine with physical access can be connected. Please note that Interface 1 by default enables a DHCP server and DNS service for ease of access to the configuration web page. This service can be disabled unchecking the “Enable DHCP Server” checkbox on the network configuration page for Interface 1.

Interface 5 is the WAN interface and is also uncontrolled.

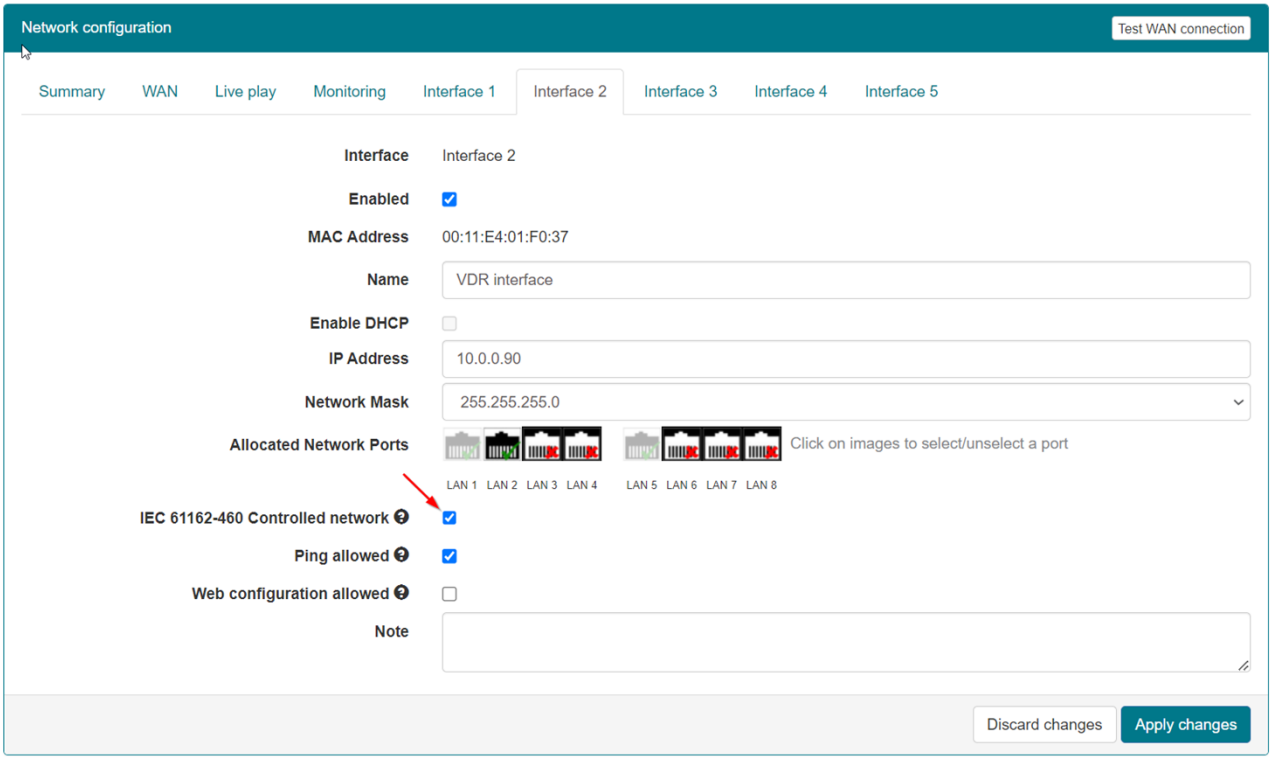


Figure 1: Configure Interface 2 as belonging to a controlled network

3.3 Compliant installation

A newly produced VRS is shipped with a compliant configuration, there are no security notices present unless the administrator or service technician actively configures the VRS improperly.

The VRS is a highly versatile product and can be configured in a vast amount of different ways, therefor it is technically possible to prepare a non-compliant configuration.

In order to assist both the service technician and the Danelec support team, non-compliant configuration entries are highlighted as 'security notices' in the configuration interface.

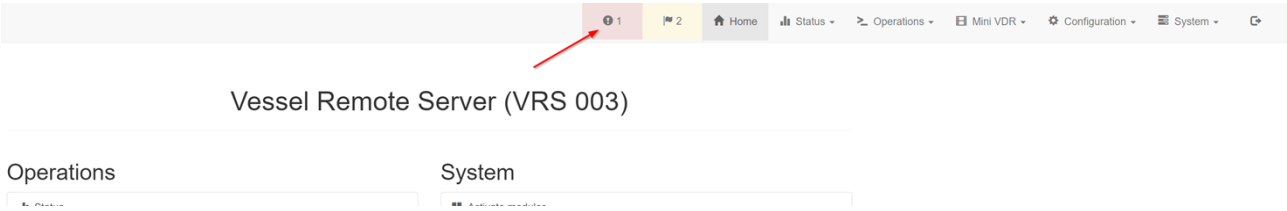


Figure 2: VRS with a security notice present

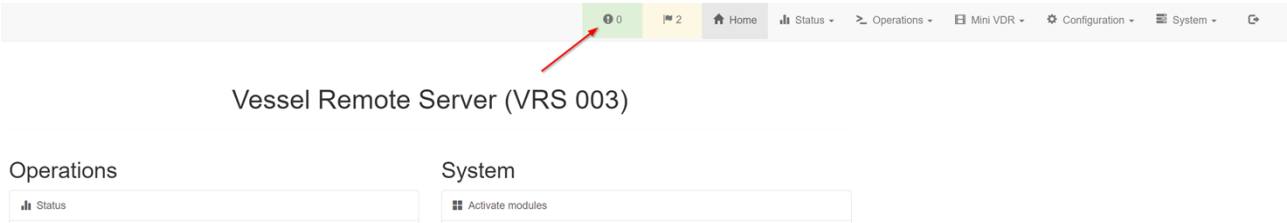


Figure 3: VRS with no security notices

Some security notices can, in specific cases, be considered too strict for that particular installation – to a high degree also depending on the remaining network topology.
That means that some security notices are more severe than others, and it may be in the ship owners interest to allow some functionality even if it carries a security notices.

On a Cloud-connected VRS, all available security notices are reviewed by the Danelec support team during commissioning.

3.3.1 Security compliance notices

Table 1 shows the complete list of security compliance notices that the VRS can present to the user along with a small explanation of the notice.

Name	Information	Explanation
Password policy	Current password policy is too lenient	The minimum compliant security policy is: - Minimum length of 8 characters - At least one digit - Require at least one upper and lower letter - At least one special character
Password too simple	Password for 'admin' user is too simple and must be changed to be compliant	Password is considered too simple and does not live up to the minimum compliant security policy.
WAN misconfiguration	WAN is setup on controlled network	The WAN interface usually provides access to the internet which by definition is not controlled.
WAN misconfiguration	VDR remote configuration service is running on WAN interface: *<interface name>*	The Onboard VDR Remote Configuration service allows access to the Danelec VDR, this should not be enabled on the WAN interface.
Remote syslog misconfiguration	Remote syslog is sent on *<interface name>*, but it is not marked as controlled network.	Remote syslopg sends information about current network connections, usually to a Network Monitoring function in the 460-Network, it should be a part of the controlled network.

Remote syslog not enabled	Remote syslog is required to be enabled in a IEC 61162-460 compliant network setting	In a IEC 61162-460 compliant network, at least one Networm Monitoring function is present, and all nodes are mandated to transmit information.
LivePlay misconfiguration	LP2a multicast configured on <*interface name*> but it is not a controlled network interface	LP2a multicast traffic is not encrypted, sensitive data can be captured. It is recommended to only use LP2a in a controlled network.
LivePlay misconfiguration	LP1 configured on <*interface name*> but it is not a controlled network interface	LP1 traffic is not encrypted, sensitive data can be captured. It is recommended to only use LP1 in a controlled network.
LivePlay misconfiguration	LP2a without TLS configured on <*interface name*> but it is not a controlled network interface	The point-to-point (TCP without TLS) version of LP2a is not encrypted. Sensitive data can be captured. It is recommended to be used only in controlled networks.
Network misconfiguration	ICMP (ping) should not be allowed on interface: <*interface name*>	There is generally no need to have ping enabled unless troubleshooting the network.
Onboard VDR configuration	Onboard VDR configuration is running on uncontrolled interface: <*interface name*>	The Onboard VDR Remote Configuration service allows access to the Danelec VDR, this should not be enabled on any uncontrolled interfaces.
Onboard VDR RMT	Onboard VDR Remote Management is running on uncontrolled interface: <*interface name*>	Onboard VDR RMT allows communication with a Danelec VDR behind the VRS, this should not be enabled on any uncontrolled interfaces.
Web interface	VRS web configuration is allowed on uncontrolled interface: <*interface name*>	The VRS web configuration interface should never be exposed towards uncontrolled networks.
FTP destination misconfigured	Destination <*destination name*> is using FTP to connect to <*host*>. FTP should be used only on a secured network, credentials and data being sent in clear text	FTP transmits username and password in cleartext and is not suited for use on or across uncontrolled networks. Use SFTP instead.
VDR connection misconfigured	VRS is connected to VDR over an uncontrolled network <*interface name*>	The VDR is inherently only meant to exist in a controlled network.
REST/JSON misconfiguration	REST/JSON destination is enabled on all network interfaces, potentially towards	REST/JSON traffic is not encrypted, sensitive data could potentially be captured. If enabled on all interfaces,

	uncontrolled networks, data is sent unencrypted	this could include uncontrolled networks.
REST/JSON misconfiguration	REST/JSON is configured on interface <*interface name*> which is not a part of controlled network, data is sent unencrypted	REST/JSON traffic is not encrypted, sensitive data could potentially be captured. It is recommended to only be used in controlled networks.
MiniVDR misconfiguration	Global MiniVDR configured on <*interface name*> but it is not a controlled network	The input to the MiniVDR is required to be coming from a controlled network but is currently configured on an uncontrolled network.
Modbus misconfiguration	Modbus channel <*modbus channel name*> is set to receive data on uncontrolled network interface: <*interface name*>	Modbus information in MiniVDR mode is required to be coming from a controlled network but is currently configured on an uncontrolled network.
Modbus misconfiguration	Modbus channel <*modbus channel name*>is set to receive data on all network interfaces, potentially including uncontrolled networks	Modbus information in MiniVDR mode is required to be coming from a controlled network but is currently configured on all interfaces which potentially include uncontrolled networks.
MiniVDR network misconfiguration	Network channel <*network channel name*> is set to receive data on all network interfaces, potentially including uncontrolled interfaces	Network channel information in MiniVDR mode is required to be coming from a controlled network but is currently configured on all interfaces which potentially include uncontrolled networks.
MiniVDR network misconfiguration	Network channel <*network channel name*> is set to receive data on uncontrolled network interface: <*interface name*>	Network channel information in MiniVDR mode is required to be coming from a controlled network but is currently configured on an uncontrolled network.
DVA misconfiguration, NMEA (UDP)	DVA streaming (UDP) traffic is not encrypted, it is sending data towards uncontrolled network through <*interface name*>	DVA data is sent unencrypted, sensitive data can be captured. It is recommended to only be used in a controlled network.
DVA misconfiguration, images	DVA streaming (UDP) traffic is not encrypted, it is sending data towards uncontrolled network through <*interface name*>	DVA data is sent unencrypted, sensitive data can be captured. It is recommended to only be used in a controlled network.
DVA misconfiguration, NMEA (TCP)	DVA streaming (TCP) accepts all connections, no IP filter is defined	If no sensitive data is sent with DVA, and it is configured towards uncontrolled networks, it is recommended to apply an IP filter so

		connection requests from unknown IP addresses are rejected.
DVA misconfiguration, NMEA (TCP)	DVA streaming (TCP) traffic is not encrypted, it is sending data towards uncontrolled network through <*interface name*>	DVA data is sent unencrypted, sensitive data can be captured. It is recommended to only be used in a controlled network.
DVA misconfiguration	DVA streaming (TCP) service is enabled on all interfaces, possibly facing uncontrolled networks	DVA data is sent unencrypted, sensitive data can be captured. It is recommended to only be used in a controlled network.
OPC UA misconfiguration	OPCUA client <*client id*> is using an uncontrolled network interface <*interface name*> with no security options	Traffic is not encrypted when no security options are enabled, this should only be used while troubleshooting.
MQTT misconfiguration	MQTT client <*client id*> is using an uncontrolled network interface: <*interface name*> with no security options	Traffic is not encrypted when no security options are enabled, this should only be used while troubleshooting.

Table 1: Description of available security compliance notices

3.3.2 Services restricted from use in uncontrolled networks

A range of services are by default, not encouraged to be used towards uncontrolled networks, these include

- LP1
- DVA Streaming
- REST/JSON
- Onboard VDR Remote Management
- Onboard VDR Configuration

The reasoning is twofold:

1. The default stance from Danelec, is that all services which do not offer encryption of the transmitted data into uncontrolled networks, are in violation of the compliance. Danelec cannot determine on behalf of the ship owner whether his data can be considered sensitive or not.
2. Onboard VDR Configuration/Management does provide encryption and integrity protection of the data being relayed, but do not provide the possibility of enabling multi-factor authentication.

However, depending on information transmitted, network topology and network elements receiving the data streams, it might be considered allowed to utilize some of these services towards uncontrolled networks anyway.

If these services are enabled towards uncontrolled networks, it results in a security notice which must be reviewed.

If the data is not considered sensitive and the ship owner accepts and understands the potential risk involved, it may be allowed on a case-by-case basis.

3.4 Security

3.4.1 Password Guidance

The default password policy for all users, can be seen by going to ***System -> Security -> Passwords policy.***

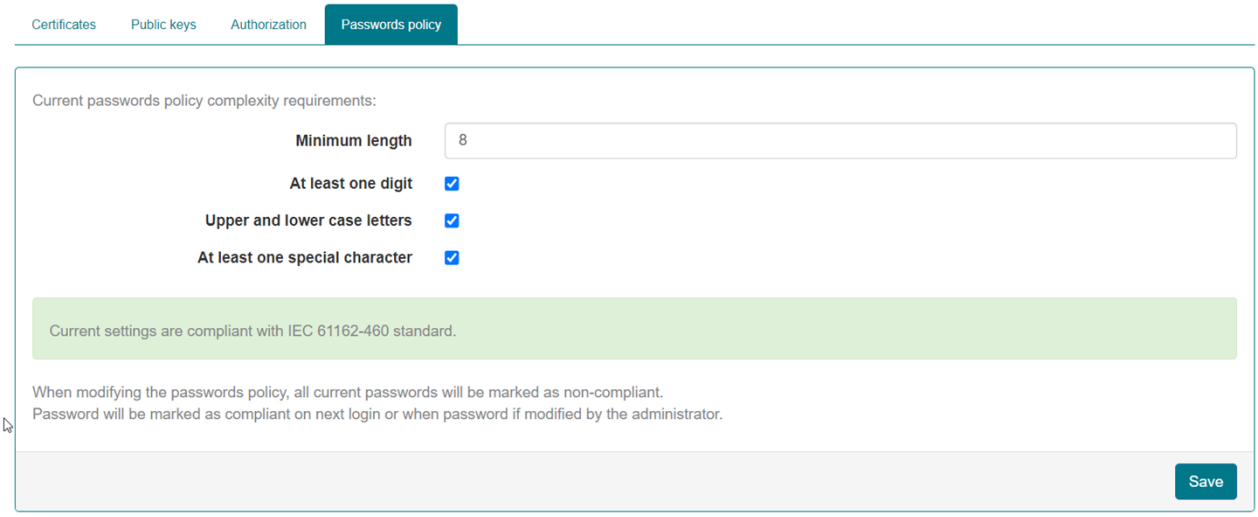


Figure 4: Minimum compliant policy settings

By default, the minimum compliant policy settings are applied (see Figure 4), these can be adjusted as needed, but relaxing the policy is not advised.

By going to ***System -> Users***, it can be seen whether the current password for each user complies with the updated password policy. The passwords for the users can be changed here as well.

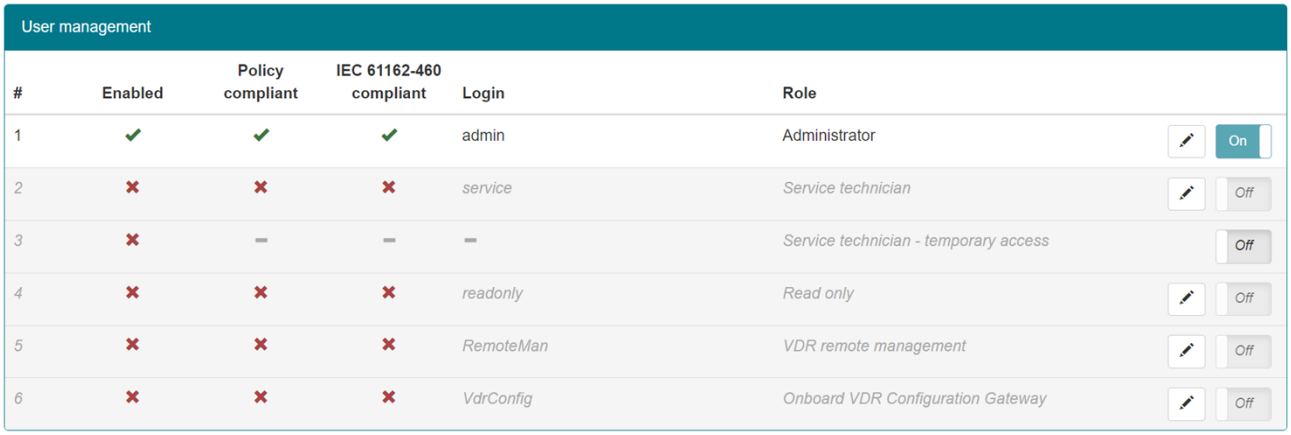


Figure 5: User management and compliance markers

When selecting a password for any user, it is highly recommended to use strong passwords. In essence, longer is stronger, and using a passphrase of random words is advisable – avoiding common words or phrases.
According to [2], a minimum of 8 characters is required, but using more is encouraged.

3.4.2 Password initialization

The following sections depicts how a VRS in both Gateway configuration and Cloud configuration is brought to a working state on board the ship, specifically it must be ensured that the password of the VRS is changed from the default password to something else – while still abiding by the password policy mentioned in 3.4.2.

Depending on the mode of operation, a different approach is followed.

3.4.2.1 Gateway VRS

A VRS in Gateway mode is intended to be used locally on the ship and not reach out to the Danelec Cloud services. In essence, the customer is responsible for setting up the VRS and most importantly making sure the default passwords are changed.

The following steps are necessary:

1. Connect to the VRS and login using the default credentials (see section 4 in [1])
2. Follow the steps in 3.4.1 to set up an appropriate password policy.
3. Follow the steps in described in section 10.1 in [1] to setup necessary users with passwords until all passwords for enabled users are marked as IEC 61162-460 compliant.

At this point normal configuration can commence.

3.4.2.2 Cloud-connected VRS

Installing a cloud-connected VRS is a two-step process, a commissioning phase managed by the service technician, which after a handover from the service technician to the Danelec support team, goes into the production phase.

The commissioning phase is a relatively short timeframe where the service technician unpacks, installs and connects the necessary cables/equipment to the VRS, according to the specifications provided by the Danelec support team in collaboration with the ship owner.

The service technician makes sure every interface is configured properly and working, and that the VRS can connect to the Cloud.

During this timeframe, the service technician manages the VRS as an administrator with the default credentials, with potential assistance from the Danelec support team.

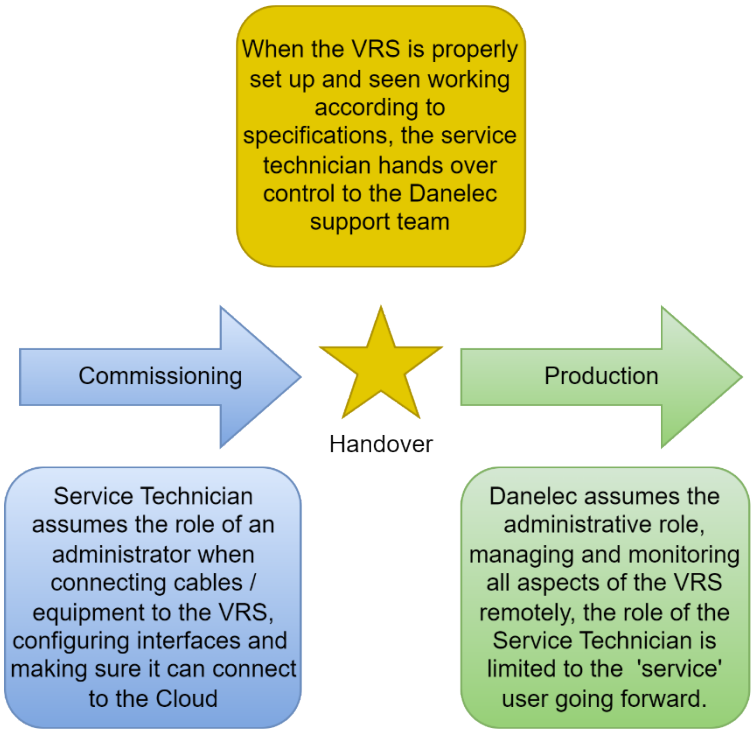


Figure 6: VRS installation from commissioning to production

Handover from the service technician to the Danelec support team is a manual process where the state of the VRS on board the vessel, changes from Commissioning to Production.

The service technician has at this point completed the installation process and the VRS is now fully controlled by the Danelec support team.

Part of the commissioning procedure involves:

- Validating that no security notices are present in the VRS

Part of the handover procedure from the Service Technician to the Danelec support team, involves:

- Changing the default password of the admin user
- Creating a 'Dump for Support' which also acts as a backup configuration in case the VRS needs to be restored.

3.4.3 Malware protection

Without physically tampering with the VRS, it is impossible to gain access to the operating system or perform any commands that may damage the unit.

No interfaces are available to perform any executable process. The firmware is digitally signed, making it impossible to update the software with anything that is not vendor-approved. If using third-party applications. They are also digitally signed by Danelec and running in a sandboxed environment, so they cannot be updated with malicious counterparts.

Continuous scanning and monitoring are not applicable for the VRS, as only applications part of the firmware are running in the VRS. It is advised to follow the guidelines for updating the VRS firmware whenever applicable to keep the VRS safe and secure.

Every user input is sanitized, client and server side for the web interface, and all uploaded configurations, remote commands are verified prior to being applied.

All the manufacturer-provided keys in the VRS are stored in a TPM chip.

3.4.4 Restricted installation instructions

The VRS device shall be installed/mounted in a location on the vessel where physical access is restricted. E.g., a safety room, alongside the VDR physical device, in a locked cabinet or other location where locked doors or other authorization mechanisms restrict physical access.

3.4.5 System use notification

When installing the VRS, a system use notification can be created.
The notification does not in itself enhance security but can be necessary for legal prosecution of violators and proving intentional breach.
The system use notification can be adjusted by shipowners as required to provide improved consistency with local laws. Policies, regulations and standards, below is an example text.

Warning: Authorized Use Only

This system is restricted to authorized users only. Unauthorized access or use is strictly prohibited and may result in disciplinary action, civil, and/or criminal penalties.

All changes or modifications to this system must be approved by the appropriate authority prior to implementation.

All login attempts and activities on this system are recorded and monitored. Unauthorized access or misuse may be subject to legal and disciplinary actions.

By proceeding, you acknowledge and agree to comply with the terms and conditions of use and understand the consequences of unauthorized changes. You are also confirming that you have obtained the required permissions for any modifications.

If you do not agree to these terms or do not have the necessary authorization, please exit immediately.

---

4 Operation and maintenance

4.1 Announcement of updates

All public releases of VRS software are accompanied by a DSI service information message, which provides the necessary documentation, including information about product security updates.
This documentation is made available to stakeholders through e-mail, and is available on-line through Danelec Partner Site and eService portal.

4.2 Backup and recovery

Backup and recovery procedures are available in different forms, this section describes how the user configuration can be backed up and stored in a secure place. It ensures that the VRS can be returned to a working state in case of service disruption or failure.

The necessary files required to restore the VRS to a known good configuration, is:

- Firmware_file.bdf
- Backup-vrs.tar.gz

The firmware file is provided by Danelec (see 4.1) while the backup file is generated once the VRS reached the desired operational state, the backup recovery procedure can also be automated by e.g. the System Management Function, see 2.3.2.

If the asset management on board the ship dictates that all recovery files should be present, the .bdf and .tar.gz can be placed on a USB stick and stored safely on board the ship, in which case the VRS can always be brought back to a working configuration, even with no access to outside networks.

4.2.1 Manual backup and recovery

In order to perform a manual backup and restore, it is necessary to connect directly to the VRS web configuration interface via LAN1, see section 4 in [1].

The firmware file needed in order to restore a VRS is not shipped with the physical VRS unit itself, it must be downloaded from either the Danelec Partner Site or eService portal.

Section 10.3.4 – 'Backup / restore configuration' in [1] describes how to create a backup of the currently running configuration.
The provided hash value should be noted down and stored alongside the backup file.

For a description on how to update the firmware of the VRS, see section 10.3.5 – 'Update firmware' in [1].

For a description on how to restore a previously saved configuration, see section 10.3.4 – 'Backup / restore configuration' in [1].

4.2.2 Automatic backup and recovery (Cloud-enabled only)

A Cloud enabled VRS is backed up in the Cloud. The Cloud management system maintains a backup of the configuration files for the connected VRS units.
From the Cloud – as long as the connection is maintained – it is possible to restore an old firmware, update to a newer firmware as well as restoring a previously working configuration.

If the VRS is not able to connect to the cloud or if a restore is to be managed offline due to the aftermath of an attack, manual intervention by a service technician is required, see 'Manual backup and recovery'.

4.3 Error logging

The VRS supports the reception of 450-compliant sentences and images when in MiniVDR mode. It is a requirement that potential errors are logged and readily available for users monitoring the data streams.
The specific errors can be seen in [4].

The error counters are present on the web interface of the VRS by going to the page ***Status -> MiniVDR***.
On this page, the configured network and image channels can be seen, along with a small icon that opens up the image and sentence processing status. The available error counters are shown below the status line (See *Figure 9: Advanced statistics for network channels* and *Figure 10: Advanced statistics for image channels* below).

Non-administrators can access these counters if the users are enabled – for example, the user "readonly" or the "service" user. See [1] for additional information on available users.

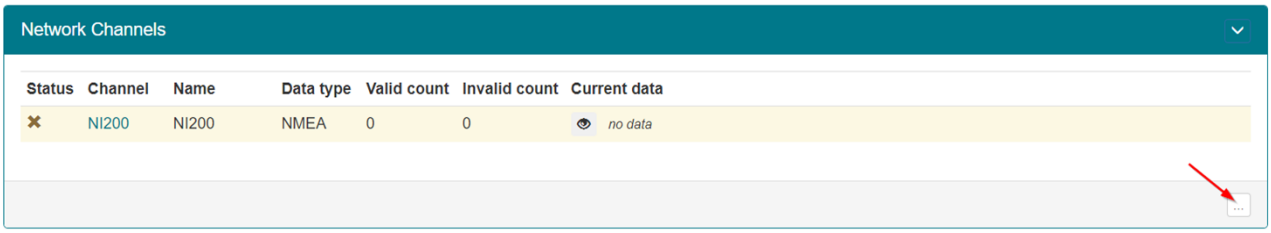


Figure 7: Advanced statistics button for network channels

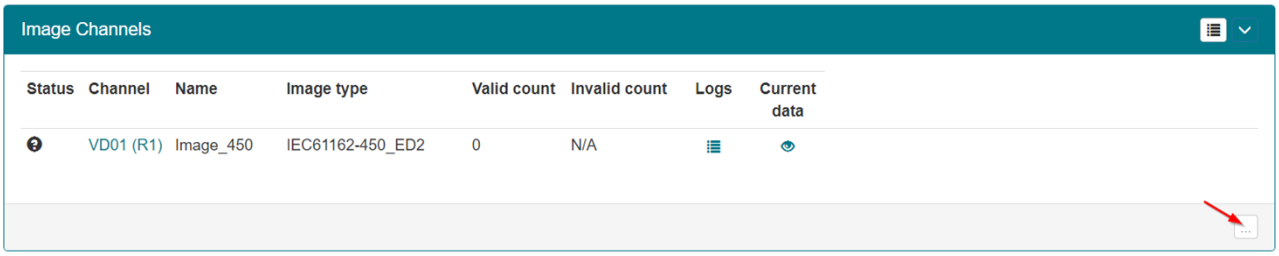


Figure 8: Advanced statistics button for image channels

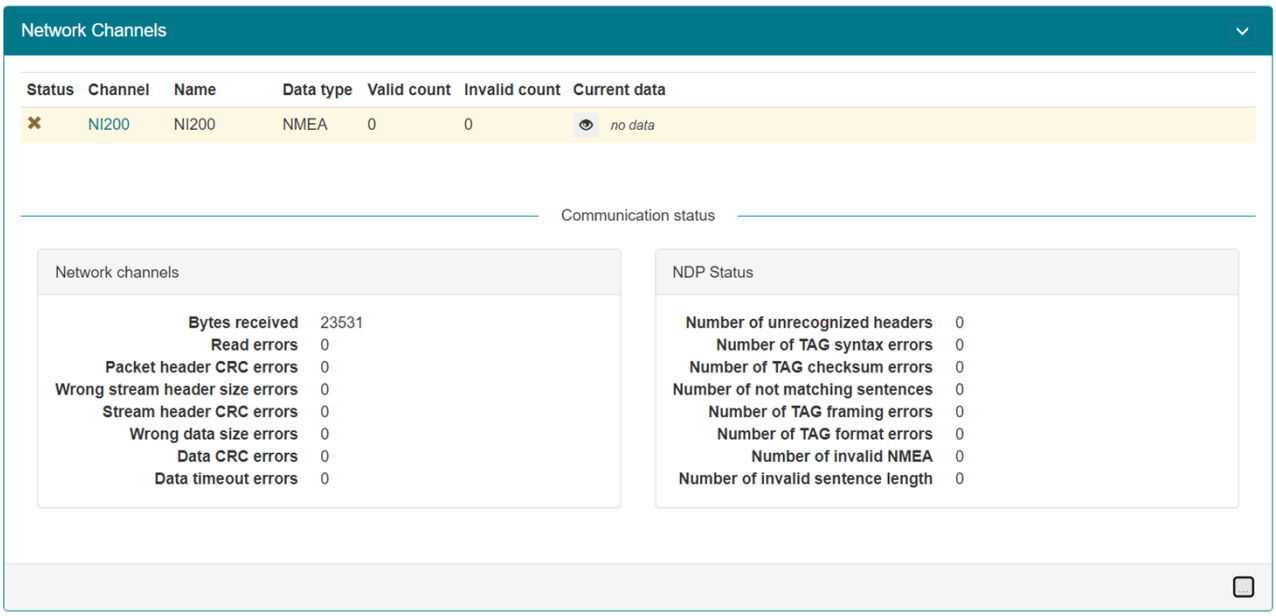


Figure 9: Advanced statistics for network channels

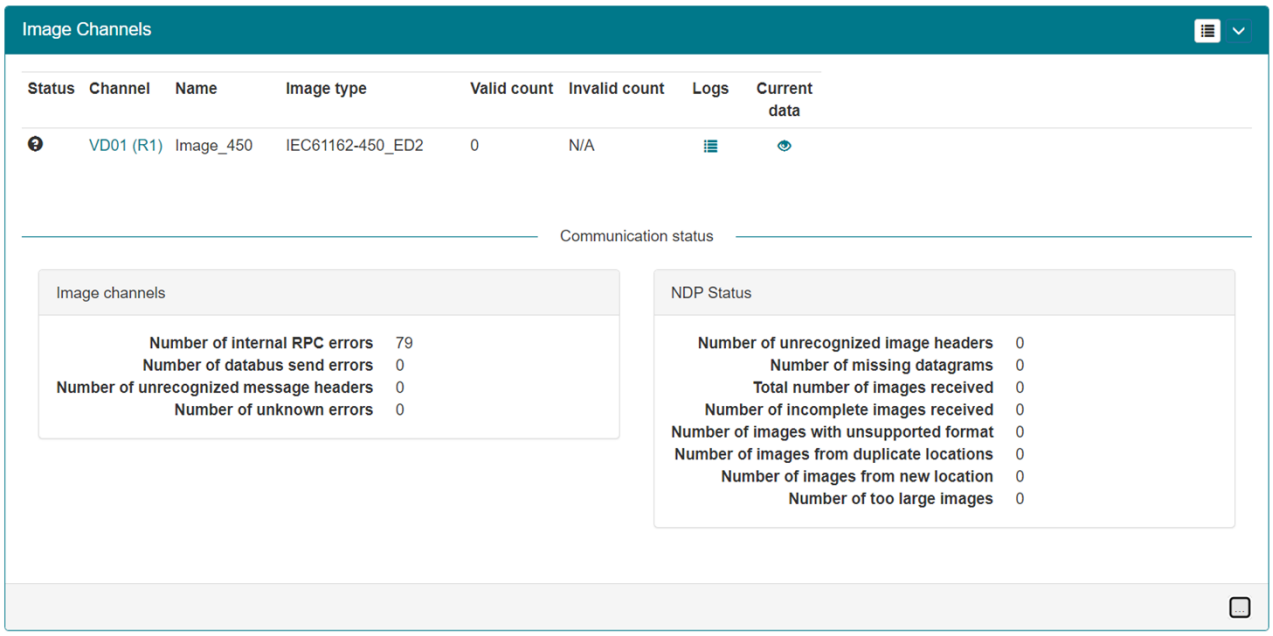


Figure 10: Advanced statistics for image channels

4.4 Network isolation

In case of a potential cyber attack, or if the networks connected to the VRS under other circumstances are required to be forcibly isolated, it is recommended that the power connector of the VRS is pulled out, leaving the VRS in powered off state - see 4.5.

The physical LAN ports on the VRS can be logically coupled in different ways when installed, depending on the specific requirements, it is therefor not possible to label specific interfaces to

unplug. The VRS is not a critical component and can safely be turned off without affecting operation of the ship – if the need arises.

4.4.1 Segregated networks

The VRS interface architecture is designed to enhance network security by isolating each interface within its own separate, non-routable broadcast domain. This isolation ensures that interfaces and networks do not route traffic between each other, thereby preventing any packet propagation across domains.

Additionally, the architecture enforces a stringent separation between uncontrolled (external) and internal networks. By maintaining this clear distinction, the system ensures that external threats are effectively contained and do not impact the internal network's operations.

4.5 Predetermined, deterministic state

The VRS provides no HMI and cannot let the crew know if it is under attack, but if by other means the crew determines that the network is under attack, the VRS can be manually powered down by unplugging the power connector.

Figure 11: Power connector for VRS 002 G2

Figure 12: Power connector for VRS 003

During an attack, the VRS will continue to provide its services in a best-effort manner (see 4.6). The VRS is not a critical component and can safely be turned off without affecting operation of the ship.

4.6 Degraded mode and lack of input

The main purpose of the VRS, is to collect data from various sensors and sources, process it and relay output to other equipment either on board the ship or on shore.

During a potential Denial of Service attack, the VRS behaves on a best-effort basis and will discard any information it cannot manage to process. it will not propagate the attack, simply cease to provide output data in a reliable fashion.

After the attack, the VRS will resume normal operation.
If necessary, the VRS can be manually forced into unpowered mode by unplugging the power, see 4.5.

If the input data streams to the VRS is suddenly stopped, it will naturally affect the output data stream(s) from the VRS, but will have no significant impact other than the data will be missing in the output data stream(s) as well.

4.7 Manual approval of external activites

A VRS in Cloud-connected mode is inherently meant to be managed by the Danelec support team from a remote location.
A VRS in Gateway mode can also be set up to receive commands from a remote location.

It is possible for a ship owner to determine which operations should be allowed, restricted by physical access or entirely forbidden both from remote locations or locally on the ship by for example a service technician, see section 'Authorization policy' in [4] for additional details.

When an operation is marked with 'Extended access required', the physical switch on the VRS must be set from "Normal operation" to Extended access" – if that is not the case, the operation will be rejected.

Figure 13: Extended access switch on VRS 002 G2

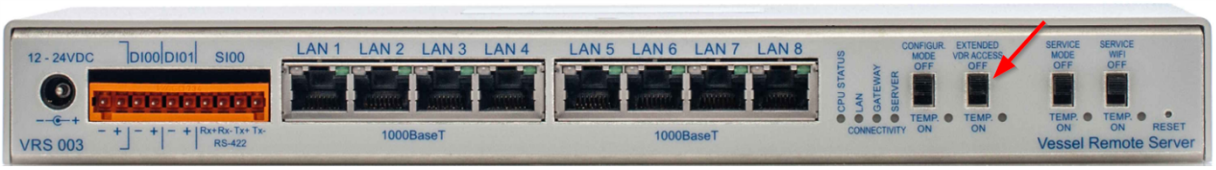


Figure 14: Extended access switch on VRS 003

4.8 Do not replace or reconfigure units

When the VRS is deployed as part of a secure IEC 61162-460 or IACS UR E27 network, any changes to both the VRS itself, or the network topology by any other than qualified professionals can compromise the integrity of the entire network and connected nodes.

In a IEC 61162-460 cyber-secure network, traffic is controlled and monitored by IEC61162-460 compliant 460-Gateways, 460-Switches and 460-Forwarders. It is imperative that no entities are added, removed, replaced or in other ways configured by other actors than the system integrator or the manufacturer themselves.

A system use notification (see 3.4.5) should provide a warning to potential intruders.

5 Technical specifications

5.1 IGMP version

The Internet Group Management Protocol version used is v3 (IGMPv3).

5.2 Hardware and wiring requirements

As per the requirements in [4], the cables utilized in the network with the VRS should be no longer than 100 meters with a minimum cable category of CAT5 STP.

5.3 Transmission rates

5.3.1 The maximum transmission rate for all supported services

The maximum transmission rates for the services available in the VRS are provided in the list below – along with the calculations if applicable.

There are a vast number of different ways the VRS can be set up and services that can be enabled in parallel. The individual services can also vary highly in transmission rates. For example, based on how many channels are enabled for sentences and how long they are.
Due to this, some assumptions have been made when gathering these numbers. The assumptions are included as part of the calculations. They can therefore be tweaked to individual installations if these assumptions do not hold up for some particular installations.

Some figures depend on the type of VRS (VRS 002 G2 or VRS 003), in which case the value for each unit is disclosed.

5.3.1.1 MODBUS

The values calculated here have been rounded upwards slightly to ensure we are well within the calculated limit. If the installation is very different, new numbers can be calculated.

Estimated sensible number of requests per second - 20
Estimated sensible number of channels - 16

Service	Calculations	Result (bits/sec)
VRS as master (UDP)	UDP single request size - 60 bytes Estimated maximum transmission rate: 60 x 20 x 16 = 19200 bytes/second	200 Kb/s
VRS as master (TCP)	TCP single request size - 78 bytes TCP single ACK size – 66 Estimated maximum transmission rate: (78 x 20 x 16) + (66 x 20 x 16) = 46080 bytes/second	400 Kb/s
VRS as slave (UDP)	UDP single response size - 301 bytes Estimated maximum transmission rate: 301 x 20 x 16 - 96320 bytes/second	800 Kb/s
VRS as slave	TCP single response size - 325 bytes	1 Mb/s

(TCP)	Estimated maximum transmission rate: 325 x 20 x 16 = 104000 bytes/second	

5.3.1.2 RDI (TFTP)

TFTP is utilized only in MiniVDR mode, with an RDI connected, and only briefly when the RDI's are booting. Subsequently, the RDIs send data to the VRS and not vice versa.

During this less than 4-second interval, a spike in data usage is seen around 14 Mb/s.

5.3.1.3 LivePlay

When the VRS is instructed to utilize LivePlay, for example, with a connected VDR Explorer, the following limits are not passed.

Service	Result (bits/sec)
LP1	3,2 Mb/s
LP2a (UDP Multicast)	2 Mb/s
LP2a (TCP)	2 Mb/s

5.3.1.4 DVA Streaming

DVA Streaming falls within the boundaries stated below.

Service	Product	Result (bits/sec)
DVA (UDP)	VRS 002 G2	5 Mb/s
DVA (TCP)	VRS 002 G2	5 Mb/s
DVA (UDP)	VRS 003	11 Mb/s
DVA (TCP)	VRS 003	11 Mb/s
DVA (UDP images)	VRS 003	3 Mb/s

Note that when combining regular DVA data with DVA Images for VRS 003, the output values are not accumulated, the maximum output value is still, due to processing limitation, at 11 Mb/s.
Images alone can utilize 3 Mb/s

5.3.1.5 Azure IoT Cloud

In cloud-controlled mode, the VRS connects to a Danelec cloud endpoint. It only applies to the WAN interface of the VRS – in the uncontrolled network.
Communication is sporadic and usually highly limited. When data is transferred from the VRS, it utilizes the bandwidth provided by the internet connection on the ship.

5.3.1.6 SFTP

SFTP data transfer off-ship only applies to the WAN interface – in the uncontrolled network. The communication depends on the specific configuration (amount of report content, number of reports, etc.)
Usually very limited but essentially utilizes the bandwidth provided by the internet connection on the ship.

5.3.1.7 Telemetry Bus

Services publishing and/or subscribing to the VRS Telemetry Bus. The limit is stated below in terms of updates per second. Multiple updates are normally bundled into a single message.

Service	Product	Limit (updates/sec)
OPC UA (Input to VRS)	VRS 002 G2	2000
OPC UA (Input to VRS)	VRS 003	5000
MQTT (Input to VRS)	VRS 002 G2	2000
MQTT (Input to VRS)	VRS 003	5000
MQTT (Output from VRS)	VRS 002 G2	2000
MQTT (Output from VRS)	VRS 003	5000

Note that multiple instances of the services can be configured. The limit above is the sum of all service instances. Depending on actual configuration, a third party system might transmit more updates per second than the limit stated above. However, a reasonable upper limit would be less than 2Mb/s for the VRS 002 G2 and less than 5Mb/s for the VRS 003.

5.3.2 450-data

The thresholds for the maximum amount of 450-compliant data under various conditions are stated below. Note that the values differ from VRS 002 G2 and VRS 003.

The required threshold values are:

1. The maximum number of datagrams per second received, intended for, and processed by the equipment
2. The maximum number of datagrams per second received by, but not intended for, the equipment
3. The maximum number of datagrams per second received by, but not intended for, the equipment at 50 % of the maximum load for item 1.

Threshold	Product	Limit (messages/sec)
1. Intended and processed datagrams	VRS 002 G2	1200
1. Intended and processed datagrams	VRS 003	1500
2. Unintended datagrams	VRS 002 G2	4200
2. Unintended datagrams	VRS 003	18000
3. 50% load of result from 1 (600 datagrams), limit of unintended datagrams	VRS 002 G2	2700
3. 50% load of result from 1 (750 datagrams), limit of unintended datagrams	VRS 003	13000

5.3.3 The maximum transmission rate for all supported VLAN

The VRS supports three VLANs (they cannot be modified) and only in its communication with the RDIs.
VLAN 769, 770, and 771, respectively, for the maximum of three RDIs.

During normal operation, the VRS does not transmit anything on these VLANs. During boot, the VRS transmits the firmware to the RDIs. This value is depicted in section 5.3.1.2.

5.4 Bandwidth

5.4.1 The maximum operational input bandwidth

When determining the maximum operation input bandwidth, the processing of incoming sentences is the limiting factor due to the available processing power. Reaching these values results in the VRS dropping some incoming packets – as described in 5.4.3

The maximum operational input bandwidth is defined as the largest amount of traffic the VRS can handle without dropping packets.

Product	Max. input bandwidth (bits/sec)
VRS 002 G2	6 Mb/s
VRS 003	10 Mb/s

5.4.2 The maximum operational output bandwidth

The VRS produces the most significant data output when the DVA module is enabled. Do note that occasional SFTP transfers can involve bandwidth usage spikes depending on the currently available bandwidth.
Also note that if RDIs are connected, an initial spike in output of TFTP traffic will occur when firmware is downloaded, as described in 5.3.1.2.

Product	Max. output bandwidth (bits/sec)
VRS 002 G2	5 Mb/s
VRS 003	11 Mb/s

5.4.3 Device behavior specified for exceeded input data rate

When the input rate described in 3.4.1 is exceeded, the VRS manages the situation by dropping the packets it does not have the resources to handle.
It affects the overall picture of the received data when the bandwidth is exceeded.

5.5 IEC 61162-450 ed. 3 cluster identifiers

IEC 61162-450 ed. 3 adds a few additional parameters to the TAG block, destination (-x) and source (-z) cluster identification.
These parameter codes are not utilized on a VRS, reception of 450-compliant data is decided only based on source / destination SFI, the cluster identification parameters are ignored.

5.6 Connection points

5.6.1 The minimum number of connection points

According to [2], the number of connection points for removable external data sources should be minimal.
VRS 002 G2 contains a USB port but is not utilized (as mentioned in [1]). It provides no connectivity to any connected equipment.

VRS 003 contains no interface for removable external data sources.

5.7 Reserved IP addresses

The VRS uses very few internal IP addresses and only if specific services are enabled.

The table below shows the IP address(es) reserved if the corresponding module is enabled.

Module (service)	IP address
RDI	192.168.238.0/24

5.8 Service list

The VRS provides various services. The table below depicts the relationship between services and the protocol and port numbers used by the services.
The direction value indicates whether traffic is inbound towards the VRS or if the VRS is transmitting.

Service	L4 Protocol	Port(s)	Direction	Classification
MiniVDR				
IEC 61162-450 Multicast	UDP	239.192.0.1:60001 – 239.192.0.16:60016	Incoming	450
IEC 61162-450 Unicast	UDP	User-defined	Incoming	450
Unicast, untagged	UDP	User-defined	Incoming	ONF
Multicast, untagged	UDP	User-defined group and port	Incoming	ONF
MODBUS Slave	UDP	User-defined	Incoming	ONF
MODBUS Master	UDP	User-defined IP and port	Outgoing	ONF
MODBUS Slave	TCP	User-defined	Listen	ONF
MODBUS Master	TCP	User-defined IP and port	Connect	ONF
RDI-S(J) (DHCP/VLAN mode)	Part of the MiniVDR configuration. 3 instances can be applied, each separated and identified by a unique VLAN (769, 770, and 771)			

DHCP	UDP	67	Incoming	ONF
DHCP	UDP	68	Outgoing	ONF
TFTP	UDP	69	Incoming / Outgoing	ONF
Data	TCP	37000	Listen	ONF
Data	TCP	2221	Listen	ONF
RDI-S(J) (Static IP mode)	Part of the MiniVDR configuration. The RDI-S(J) is identified via its static IP address			
HTTP	TCP	2280	Listen	ONF
Data	TCP	37000	Listen	ONF
Data	TCP	2221	Listen	ONF
Live Client / Server	Various versions of the Danelec streaming protocol. The VRS acts as a Live Client towards the VDR and a Live Server when (re)-playing data			
LP1	UDP	65000 - 65010	Incoming / Outgoing	ONF
LP1	UDP	65000 - 65001	Incoming / Outgoing	ONF
LP2a	TCP	64000	Listen	ONF
LP2a	UDP Multicast	239.192.1.111:64001 - 239.192.1.115:64005	Outgoing	ONF
LP2a	UDP Multicast	239.192.1.101:64001 - 239.192.1.105:64005	Incoming	ONF
LP2b	TCP	443	Connect	ONF
DVA streaming				
Images	UDP	User-defined (default 22000)	Outgoing	ONF
Serial etc.	TCP/UDP	User-defined (default 23000)	Listen	ONF
FTS	File Transport System.			
SFTP	TCP	User-defined (default 22)	Listen / Connect	ONF
FTS configuration	TCP	User-defined (default 4000)	Listen	ONF
Misc				
REST/JSON interface	TCP	User-defined (Default 8080)	Listen	ONF
Configuration interface	TCP	443	Listen	ONF
Azure IoT Cloud	TCP	443	Connect	ONF

- MQTT over TLS	TCP	8883	Listen	ONF
DNS	UDP	53	Outgoing / Incoming	ONF
Transfer Controller	TCP	User-defined (default 22)	Connect	ONF
MQTT	TCP	User-defined (default 8883)	Connect	ONF
OPC UA	TCP	User-defined (default 4840)	Connect	ÓNF
NavBox	TCP	User-defined (default 8010)	Connect	ONF
VDR Management				
Remote configuration	TCP	80, 44300	Listen	ONF
Remote management	TCP	81, User-defined	Listen	ONF

5.9 Use of cryptography

The VRS system has a wide range of services which apply the use of cryptography to protect both the confidentiality as well as integrity of the information both received and transmitted in all aspects of the application.

A dedicated TPM chip is used for secure storage of cryptographic keys as well as attestation service when communicating with the Cloud.

Encryption / decryption is implemented using OpenSSL or the TPM where applicable.

All services by default approved for communication with uncontrolled networks, utilize mTLS, TLS or other means of encryption, implemented using X.509 certificates.

All keys utilized by the VRS are at least 2048 bits long.
Hash functions for all certificates is SHA-256, which is also predominantly used elsewhere, a few checksums utilize SHA-1.

Firmware and potential hosted applications are digitally signed by Danelec and the VRS will reject running firmware or applications that cannot be verified.

6 Appendix A

Example of output from the VRS when monitoring is enabled, and the network status is transmitted to a remote syslog.

```
<13>1 2023-06-07T09:24:18.279259+00:00 - 450-onf - - -  Network configuration changed
<13>1 2023-06-07T09:24:18.281285+00:00 - 450-onf - - -  Controlled network interface: Interface 2
<13>1 2023-06-07T09:24:18.283135+00:00 - 450-onf - - -  Controlled network interface: Interface 3
<13>1 2023-06-07T09:24:18.284954+00:00 - 450-onf - - -  Controlled network interface: Interface 4
<13>1 2023-06-07T09:24:18.301114+00:00 - 450-onf - - -  live_client, LP1, analog/digital, udp | Interface 2, input (b/s): 0, output
(b/s): 0
<13>1 2023-06-07T09:24:18.304278+00:00 - 450-onf - - -  live_client, LP1, audio, udp | Interface 2, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.307179+00:00 - 450-onf - - -  live_client, LP1, command, udp on 65001 | Interface 2, input (b/s): 0, output
(b/s): 0
<13>1 2023-06-07T09:24:18.310045+00:00 - 450-onf - - -  live_client, LP1, connect, udp on 65000 | Interface 2, input (b/s): 0, output
(b/s): 0
<13>1 2023-06-07T09:24:18.312946+00:00 - 450-onf - - -  live_client, LP1, image, udp | Interface 2, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.315929+00:00 - 450-onf - - -  live_client, LP1, log, udp | Interface 2, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.318711+00:00 - 450-onf - - -  live_client, LP1, serial, udp | Interface 2, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.321704+00:00 - 450-onf - - -  liveplay, LP1, analog/digital, udp | Interface 2, input (b/s): 0, output
(b/s): 0, maximum transfer rate: 1.2 MB/s
<13>1 2023-06-07T09:24:18.324573+00:00 - 450-onf - - -  liveplay, LP1, audio, udp | Interface 2, input (b/s): 0, output (b/s): 0,
maximum transfer rate: 1.2 MB/s
<13>1 2023-06-07T09:24:18.327388+00:00 - 450-onf - - -  liveplay, LP1, command, udp | Interface 2, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.330218+00:00 - 450-onf - - -  liveplay, LP1, connect, udp | Interface 2, input (b/s): 0, output (b/s): 8
<13>1 2023-06-07T09:24:18.333147+00:00 - 450-onf - - -  liveplay, LP1, image, udp | Interface 2, input (b/s): 0, output (b/s): 0,
maximum transfer rate: 1.2 MB/s
<13>1 2023-06-07T09:24:18.335981+00:00 - 450-onf - - -  liveplay, LP1, log, udp | Interface 2, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.339070+00:00 - 450-onf - - -  liveplay, LP1, serial, udp | Interface 2, input (b/s): 0, output (b/s): 0,
maximum transfer rate: 1.2 MB/s
<13>1 2023-06-07T09:24:18.342023+00:00 - 450-onf - - -  liveplay, LP2a, TCP playback, tcp | Interface 2, input (b/s): 0, output
(b/s): 0, maximum transfer rate: 1.65 MB/s
<13>1 2023-06-07T09:24:18.344965+00:00 - 450-onf - - -  rsyslogd, udp to 10.1.0.42/32 on 5140 | any, input (b/s): 0, output (b/s): 0
<13>1 2023-06-07T09:24:18.348098+00:00 - 450-onf - - -  vdr_config, api, tcp to 10.0.0.100/32 on 81 | Interface 2, input (b/s): 0,
output (b/s): 0
<13>1 2023-06-07T09:24:18.350961+00:00 - 450-onf - - -  vdr_config, vdr, tcp to 10.0.0.100/32 on 80 | Interface 2, input (b/s): 0,
output (b/s): 0
<13>1 2023-06-07T09:24:18.353810+00:00 - 450-onf - - -  web_server, https, tcp | Interface 1, input (b/s): 8, output (b/s): 10
```